

Day 2: Arriaga decomposition

José Manuel Aburto and Ana Cristina Gómez
Ugarte Valerio

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Recap

- ▶ We decompose to better understand aggregate measures changes/differences
- ▶ Decomposition methods improve on existing standardisation methods

Kitagawa decomposition

Decomposing a weighted mean into

- ▶ Difference in rates
- ▶ Difference in population composition (weight)
 - can be along any categorical dimension
 - age, marital status, gender, education

Quiz time!

Kitagawa decomposition: extensions

(Kitagawa-)Oaxaca-Blinder decomposition method

- ▶ Separately proposed by Ronald Oaxaca and Alan Blinder in 1973
- ▶ Generalisation of Kitagawa method (Oaxaca, 2025)
- ▶ Regression-based: it can include multiple independent variables

Kitagawa decomposition: extensions

Das Gupta decomposition method

- ▶ Proposed by Prithwis Das Gupta in 1978
- ▶ Kitagawa can be used with more than one compositional factor, but this creates interaction terms
- ▶ Das Gupta eliminates these terms by equally redistributing them across the main terms (symmetric approach)

Kitagawa decomposition: extensions

Chevan and Sutherland decomposition method (extension of Das Gupta's method)

- ▶ Proposed by Albert Chevan and Micheal Sutherland in 2009
- ▶ Estimates contributions of specific categories vs of overall variables (e.g. married individuals vs marital status)
- ▶ Apply decomposition to polytomous dependent variable

What about other measures?

- ▶ Kitagawa: analytical decomposition of a weighted mean
- ▶ One can do that with more aggregate measures
- ▶ For example, life expectancy

Eduardo E. Arriaga

- ▶ Spanish-born, Argentinian and US demographer, worked in US and Argentinian universities
- ▶ Mostly worked on mortality, also fertility, population change and urbanisation
- ▶ Focus on Latin America
- ▶ *Measuring and explaining the change in life expectancies*
1984
→ similar approaches developed by other researchers at the same time



Arriaga decomposition

Decompose difference in life expectancy by age

Total change is the sum of age-specific changes:

$$\Delta LE = \sum_x ({}_n\Delta_x)$$

Arriaga decomposition

$${}_n\Delta_x = \underbrace{\frac{l_x^1}{l_0^1} \left(\frac{{}_nL_x^2}{l_x^2} - \frac{{}_nL_x^1}{l_x^1} \right)}_{\text{Direct effect}} +$$

$$\underbrace{\frac{T_{x+n}^2}{l_0^1} \left(\frac{l_x^1}{l_x^2} - \frac{l_{x+n}^1}{l_{x+n}^2} \right)}_{\text{Indirect and interaction effects}}$$

Change in mortality rates between ages x and $x + n$ affects

- ▶ Number of years lived between ages x and $x + n \rightarrow$ direct effect
- ▶ Number of years lived after age $x + n$ by new survivors $\rightarrow$ indirect and interaction effect

Arriaga decomposition

$${}_n\Delta_x = \underbrace{\frac{l_x^1}{l_0^1} \left(\frac{{}_nL_x^2}{l_x^2} - \frac{{}_nL_x^1}{l_x^1} \right)}_{\text{Direct effect}} +$$

$$\underbrace{\frac{T_{x+n}^2}{l_0^1} \left(\frac{l_x^1}{l_x^2} - \frac{l_{x+n}^1}{l_{x+n}^2} \right)}_{\text{Indirect and interaction effects}}$$

Change in mortality rates between ages x and $x + n$ affects

- ▶ Number of years lived between ages x and $x + n \rightarrow$ direct effect
- ▶ Number of years lived after age $x + n$ by new survivors $\rightarrow$ indirect and interaction effect

For open-ended age-group there is only a direct effect:

$${}_{\infty}\Delta_x = \frac{l_x^1}{l_0^1} \left(\frac{T_x^2}{l_x^2} - \frac{T_x^1}{l_x^1} \right)$$

An example

PNAS

RESEARCH ARTICLE

DEMOGRAPHY

 OPEN ACCESS

Significant impacts of the COVID-19 pandemic on racial and ethnic differences in US mortality

José Manuel Aburto^{a,b,c,1} , Andrea M. Tilstra^{a,d,1}, Ginevra Floridi^a , and Jennifer Beam Dowd^a 

Edited by Douglas Massey, Princeton University, Princeton, NJ; received April 2, 2022; accepted June 20, 2022

An example

Let's try ourselves

Arriaga decomposition by causes of death

We can extend this method to account for contributions from different causes of death i :

$$\begin{aligned} {}_n\Delta_x^i &= {}_n\Delta_x \frac{{}_nm_{x,i}^2 - {}_nm_{x,i}^1}{{}_nm_x^2 - {}_nm_x^1} \\ &= {}_n\Delta_x \frac{{}_nR_{x,i}^2({}_nm_x^2) - {}_nR_{x,i}^1({}_nm_x^1)}{{}_nm_x^2 - {}_nm_x^1} \end{aligned}$$

Where

- ▶ ${}_nR_{x,i}^t$ is the proportion of deaths from cause i in ages x to $x + n$ and population t
- ▶ ${}_n\Delta_x$ is the contribution of all-cause mortality differences in the same age group
- ▶ ${}_nm_x^t$ is the mortality rate in the same age group of population t

Arriaga decomposition by causes of death

All-cause contribution, weighted by the change in cause-specific mortality as a proportion of change in all-cause mortality

An example

Back to R!

Kitagawa-Arriaga combination

New method to decompose differences in life expectancy by population composition

- ▶ Proposed by Riffe, Tursun-Zade and Trias Llimós in 2024 (working paper)
- ▶ Kitagawa decomposition to distinguish e_x (rate) vs composition effects
- ▶ Arriaga decomposition of differences in e_x
- ▶ Rescaling to make everything match

References

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- A. Chevan and M. Sutherland (2009). "Revisiting Das Gupta: Refinement and extension of standardization and decomposition", *Demography*, 46: 429—449.
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